The slope of the normal is the negative of the reciprocal of the tangent which in this case is $\frac{\sin 3t}{\cos 2t}$. Hence, the equation of the normal is

$$y - 3\sin 2t = \frac{\sin 3t}{\cos 2t}(x - 2\cos 3t).$$

That is, $x\sin 3t - y\cos 2t = 2\sin 3t\cos 3t - 3\sin 2t\cos 2t = \sin 6t - \frac{3}{2}\sin 4t$.

7.2.5 Angle between two curves

Definition 7.3

Angle between two curves, if they intersect, is defined as the acute angle between the tangent lines to those two curves at the point of intersection.

For the given curves, at the point of intersection using the slopes of the tangents, we can measure the acute angle between the two curves. Suppose $y = m_1x + c_1$ and $y = m_2x + c_2$ are two lines, then the acute angle θ between these lines is given by,

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right| \qquad \ldots (3)$$

where m_1 and m_2 are finite.

Remark

(i) If the two curves are parallel at (x_1, y_1), then $m_1 = m_2$.

(ii) If the two curves are perpendicular at (x_1, y_1) and if m_1 and m_2 exists and finite then $m_1m_2 = -1$.

Example 7.14

Find the angle between $y = x^2$ and $y = (x-3)^2$.

Solution

Let us now find the point of intersection of the two given curves. Equating $x^2 = (x-3)^2$ we get, $x = \frac{3}{2}$. Therefore, the point of intersection is $\left(\frac{3}{2}, \frac{9}{4}\right)$. Let θ be the angle between the curves. The slopes of the curves are as follows :

For the curve $y = x^2$,

$$\frac{dy}{dx} = 2x.$$

$$\text{Let } m_1 = \frac{dy}{dx} \text{ at } \left(\frac{3}{2}, \frac{9}{4}\right) = 3.$$

For the curve $y = (x-3)^2$,

$$\frac{dy}{dx} = 2(x-3).$$

$$\text{Let } m_2 = \frac{dy}{dx} \text{ at } \left(\frac{3}{2}, \frac{9}{4}\right) = -3.$$

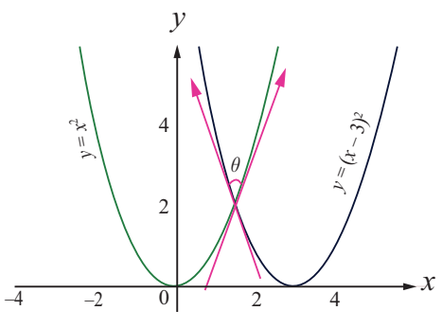


Fig.7.10

12th_Maths_Vol 2_EM_CH 7_Differential Calculus.indd 11 16-12-2022 16:24:48